

# Notes: Chen, Singal, and Whitelaw (JFE, 2016)

## Comovement Revisited

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# 1 Big picture

- **Question.** Is the documented “excess comovement” around S&P 500 additions and stock splits really evidence of nonfundamental asset-class demand, or can it be explained by more standard beta dynamics?
- **Background.** Barberis, Shleifer, and Wurgler (2005) argue that stocks added to the S&P 500 begin to comove more with the S&P 500 and less with nonindex stocks. Green and Hwang (2009) argue that stock splits generate similar style-based comovement by price group.
- **Core critique.** The bivariate regressions used in the prior literature do not measure the economic magnitude of excess comovement cleanly. Their coefficients can move for reasons unrelated to nonfundamental demand.
- **Main methodological point.** A univariate regression on the old group is also informative. If a stock genuinely moves from one asset class to another, its loading on the old group should fall and its loading on the new group should rise.
- **Initial replication.** The authors replicate and extend the prior results. Bivariate regressions often look like excess comovement, but univariate regressions are less supportive.
- **Key alternative explanation.** Stocks added to the S&P 500 and stocks that split are typically past winners. Momentum winners experience increasing market betas around the portfolio formation/event date.
- **Matched-sample design.** The authors match event stocks to nonevent stocks with similar size and past returns. If comovement results are due to momentum-driven beta changes, the matched stocks should show similar coefficient changes even though they do not switch groups.
- **Main empirical result.** After matching on size and momentum, almost all evidence of excess comovement disappears for both S&P 500 additions and stock splits.
- **Dimson adjustment.** For S&P 500 additions, nonsynchronous trading and microstructure effects matter because added firms can be smaller relative to S&P 500 constituents. Dimson-adjusted betas further weaken the excess-comovement interpretation.
- **Economic takeaway.** Apparent comovement is largely generated by time-varying betas and sample selection into past winners, not by economically large nonfundamental asset-class demand.

# 2 Data and measurement

## 2.1 S&P 500 additions

- The S&P 500 addition sample covers 1976–2012.
- The pre-event period is a one-year window ending at the end of the month before the announcement.
- The post-event period is a one-year window beginning the month after the effective date of index inclusion.
- Group 1 is the non-S&P 500 portfolio; group 2 is the S&P 500 portfolio.
- The sample is divided into 1976–1987, 1988–2000, 2001–2012, and the full 1976–2012 period.

**Interpretation:** S&P 500 additions are useful because earlier literature interprets changes in loadings as evidence that an index membership shock creates new nonfundamental comovement with the index.

## 2.2 Stock splits

- The stock split sample follows Green and Hwang’s two-for-one split design and is extended to 2012.
- A stock enters the sample if its presplit price is at least \$10.
- The old group is the high-priced portfolio: stocks with prices within 25% of the presplit price.
- The new group is the low-priced portfolio: stocks with prices within 25% of the postsplit price.
- Regression windows are again one year before and one year after the split event.

**Interpretation:** Splits are meant to be useful because price-group membership changes without obvious direct changes to fundamentals. The paper asks whether the apparent style shift is instead caused by beta dynamics of winner stocks.

## 2.3 Momentum portfolios

- Momentum portfolios are formed each June using the previous 252 trading days of returns, with a skip month.
- Stocks with price at least \$10 and outside the bottom NYSE size decile are sorted into ten momentum deciles.
- Winner stocks are the top decile; loser stocks are the bottom decile.
- Betas are estimated over rolling 252-day windows from two years before portfolio formation to two years after.

**Interpretation:** Momentum is not only a return characteristic in this paper. It is also a beta-change characteristic: winner stocks tend to experience rising market betas around the formation period.

## 2.4 Matched samples

- For each event stock, the authors select a nonevent stock from the same size decile with similar past-year return.
- Matched stocks do not join the S&P 500 and do not split, but they resemble the event stocks in size and momentum.
- The main test compares beta changes for the event sample against beta changes for matched stocks.

**Interpretation:** If the matched sample shows the same beta shifts as event stocks, then the original comovement evidence is likely driven by momentum-related beta changes rather than group membership itself.

## 2.5 Dimson adjustment

- The paper also estimates Dimson-style betas using leads and lags of group returns.
- This adjustment addresses nonsynchronous trading and microstructure effects.
- It is particularly important for S&P 500 additions, where added stocks can be smaller than existing index stocks.

**Interpretation:** Dimson adjustment prevents stale-price or delayed-response effects from being misinterpreted as new comovement with the S&P 500.

# 3 Models and empirical specifications

## 3.1 Return decomposition

The model has a stock  $y$  that moves from group 1 to group 2. Returns are written as

$$y_t = b_{yt}f_t + c_{1t}u_{1t} + c_{2t}u_{2t} + e_{yt}, \quad (1)$$

$$x_{1t} = b_{1t}f_t + u_{1t} + e_{1t}, \quad (2)$$

$$x_{2t} = b_{2t}f_t + u_{2t} + e_{2t}. \quad (3)$$

Here  $f_t$  is a common fundamental factor,  $u_{1t}$  and  $u_{2t}$  are nonfundamental group-specific shocks, and  $e_{yt}, e_{1t}, e_{2t}$  are idiosyncratic shocks.

**Interpretation:** The model separates fundamental beta changes from nonfundamental group-specific comovement. This distinction is the paper's main conceptual contribution.

### 3.2 Orthogonality assumptions

The paper assumes

$$\text{cov}(u_{1t}, u_{2t}) = 0, \quad (4)$$

$$\text{cov}(u_{it}, f_t) = 0, \quad (5)$$

$$\text{cov}(u_{it}, e_{jt}) = 0. \quad (6)$$

**Interpretation:** These assumptions make it possible to interpret coefficient changes as coming from fundamental betas, group-specific shocks, or volatility changes.

### 3.3 Excess comovement hypothesis

If stock  $y$  moves from group 1 to group 2, the excess-comovement hypothesis implies

$$c_{1t} = c_1 > 0, \quad c_{2t} = 0, \quad \bar{c}_{1t} = 0, \quad \bar{c}_{2t} = \bar{c}_2 > 0. \quad (7)$$

**Interpretation:** The economic content of excess comovement is not merely that regression coefficients move. It is that the stock stops loading on the old nonfundamental group shock and starts loading on the new nonfundamental group shock.

### 3.4 Economic magnitude of excess comovement

A natural measure of excess comovement is the fraction of the stock's variance due to the relevant nonfundamental group shock:

$$\frac{c_1^2 \sigma_{u1}^2}{\sigma_y^2} \quad \text{and} \quad \frac{\bar{c}_2^2 \bar{\sigma}_{u2}^2}{\bar{\sigma}_y^2}. \quad (8)$$

**Interpretation:** This measure is an  $R^2$ -type object. Regression coefficients alone are not enough because they do not reveal how large the nonfundamental shocks are.

### 3.5 Regression specifications

The prior literature focuses on three regressions, estimated before and after the event:

$$y_t = \alpha + \beta_1 x_{1t} + \epsilon_t, \quad (9)$$

$$y_t = \alpha + \beta_2 x_{2t} + \epsilon_t, \quad (10)$$

$$y_t = \alpha + \beta_{1b} x_{1t} + \beta_{2b} x_{2t} + \epsilon_t. \quad (11)$$

- The first univariate regression loads on the old group.
- The second univariate regression loads on the new group.
- The bivariate regression loads on both groups.

**Interpretation:** Prior papers mainly emphasized the new-group coefficient and the bivariate regression. Chen, Singal, and Whitelaw emphasize that the old-group univariate coefficient is crucial for interpretation.

### 3.6 Implications for coefficient changes

Under the excess-comovement hypothesis, a move from group 1 to group 2 should produce

$$\Delta\beta_1 < 0, \quad \Delta\beta_2 > 0. \quad (12)$$

However, the paper shows that bivariate coefficients can be sensitive to other parameters, including:

- changes in the stock's fundamental beta;
- changes in group-level idiosyncratic volatility;
- changes in group fundamental betas;
- nonsynchronous trading effects.

**Interpretation:** The same coefficient pattern can arise with no true excess comovement. This is why the matched-sample and Dimson-adjusted tests are necessary.

### 3.7 Matched-sample difference-in-differences logic

Let  $\Delta\beta^{sample}$  be the beta change for event stocks and  $\Delta\beta^{match}$  be the beta change for matched stocks. The key comparison is

$$\Delta\Delta\beta = \Delta\beta^{sample} - \Delta\beta^{match}. \quad (13)$$

**Interpretation:** If  $\Delta\Delta\beta$  is small, the event did not create special comovement. The beta change is instead a feature of stocks with similar momentum and size.

## 4 Identification and empirical design

### 4.1 Why the original evidence is not enough

- Bivariate coefficients can move even when the stock has no new nonfundamental group loading.
- Changes in fundamental beta can make both old-group and new-group univariate coefficients rise.
- Group-level volatility changes can alter bivariate coefficients mechanically.
- Therefore, coefficient changes must be interpreted through a model, not as direct evidence of excess comovement.

**Interpretation:** The paper’s empirical design is motivated by a measurement critique: the original regressions do not isolate nonfundamental comovement.

### 4.2 Why momentum matters

- S&P 500 additions tend to be successful firms with high prior returns.
- Stocks that split are also strong past winners.
- Momentum winners have rising market betas around the formation and holding periods.
- This beta pattern can mimic the appearance of greater comovement with the group the stock is joining.

**Interpretation:** Momentum creates a selection problem. Event stocks are not random; they are beta-changing winners.

### 4.3 Why matched samples matter

- Matching on size and past returns selects nonevent stocks with similar beta dynamics.
- These matched stocks do not change groups, so any similar beta movement cannot be caused by index inclusion or stock-split price-group movement.
- If matched stocks display similar regression coefficient changes, excess comovement is not the main explanation.

**Interpretation:** The matched-sample test is the paper’s most direct empirical rebuttal of the prior comovement interpretation.

### 4.4 What the paper cannot fully prove

- The model is stylized and does not fully decompose every source of covariance.
- Matching on size and momentum may not capture every relevant stock characteristic.
- The tests focus on index additions and stock splits, two important but specific settings.
- The paper weakens the evidence for excess comovement but does not show that nonfundamental demand never matters.

**Interpretation:** The conclusion is careful: almost all evidence of economically large excess comovement disappears in these settings once beta dynamics and matched controls are used.

## 5 Tables and figures

The tables and figure below are directly cropped from the paper and grouped compactly.

### 5.1 Table 1: S&P 500 additions

Read:

- Panel A shows univariate regressions for additions. The loading on the S&P 500 rises after inclusion, but the loading on non-S&P stocks does not fall much.
- For 1988–2000, the S&P 500 univariate coefficient rises by 0.214, while the non-S&P coefficient rises slightly by 0.015.
- Panel B shows stronger bivariate evidence: the S&P coefficient rises and the non-S&P coefficient falls.
- The bivariate difference-of-differences is large, especially in 1988–2000 and for the full sample.

**Economic message:** Table 1 replicates the classic evidence: bivariate regressions look like excess comovement, but univariate regressions are less clean.

### 5.2 Table 2: Stock splits

Read:

- Panel A shows that split stocks’ loadings rise on both high-priced and low-priced groups after splits.
- The loading on the low-priced group rises, consistent with the prior price-category story, but the loading on the old high-priced group also rises.
- Panel B again looks more supportive of excess comovement because the bivariate loading on the low-priced group rises more.
- For the full 1976–2012 sample, the univariate difference-of-differences is only 0.013, while the bivariate difference-of-differences is 0.265.

**Economic message:** Table 2 is the first warning sign: if stock splits cause true movement from one price category to another, the old-group loading should fall, but it does not.

**Table 1**  
S&P additions.  
We estimate the univariate and bivariate regressions  
 $y_t = \alpha + \beta_1 x_{1t} + \varepsilon_t$   
 $y_t = \alpha + \beta_2 x_{2t} + \varepsilon_t$   
 $y_t = \alpha + \beta_{12} x_{1t} + \beta_{22} x_{2t} + \varepsilon_t$   
for a sample of stocks that are added to the S&P500 index from 1976 through 2012. The pre-event estimation period covers a one-year window ending at the end of the month preceding announcement, while the postevent period covers the one-year window starting the month after the effective date of index change.  $x_{1t}$  and  $x_{2t}$  are returns to non-S&P500 index and S&P500 index at time  $t$ . Panel A reports the univariate regression results, and Panel B reports the bivariate regression results. In each cell, the first number is the mean, and the second number is the corresponding  $t$ -statistic, where standard errors are clustered by month.

| Panel A: Univariate regressions |      |                 |                 |                        |                 |                 |                                               |                  |
|---------------------------------|------|-----------------|-----------------|------------------------|-----------------|-----------------|-----------------------------------------------|------------------|
| Non-S&P500                      |      |                 | S&P500          |                        |                 | Diff. of diff.  |                                               |                  |
| Sample period                   | nobs | $\hat{\beta}_1$ | $\hat{\beta}_1$ | $\Delta \hat{\beta}_1$ | $\hat{\beta}_2$ | $\hat{\beta}_2$ | $\Delta \hat{\beta}_2 - \Delta \hat{\beta}_1$ |                  |
| 1976–1987                       | 197  | 1.271<br>29.422 | 1.313<br>27.202 | 0.042<br>1.055         | 0.962<br>24.643 | 1.024<br>26.830 | 0.062<br>2.305                                | 0.020<br>0.763   |
| 1988–2000                       | 269  | 1.263<br>29.839 | 1.278<br>27.503 | 0.015<br>0.313         | 0.984<br>24.669 | 1.198<br>23.830 | 0.214<br>6.243                                | 0.199<br>4.938   |
| 2001–2012                       | 214  | 1.050<br>30.548 | 1.125<br>28.736 | 0.075<br>2.496         | 1.086<br>27.526 | 1.157<br>35.741 | 0.071<br>2.439                                | –0.004<br>–0.137 |
| 1976–2012                       | 680  | 1.198<br>49.851 | 1.240<br>46.610 | 0.042<br>1.734         | 1.010<br>43.566 | 1.134<br>44.294 | 0.125<br>6.556                                | 0.083<br>4.080   |

  

| Panel B: Bivariate regressions |      |                    |                    |                           |                    |                    |                                                     |                 |
|--------------------------------|------|--------------------|--------------------|---------------------------|--------------------|--------------------|-----------------------------------------------------|-----------------|
| Non-S&P500                     |      |                    | S&P500             |                           |                    | Diff. of diff.     |                                                     |                 |
| Sample period                  | nobs | $\hat{\beta}_{1a}$ | $\hat{\beta}_{1b}$ | $\Delta \hat{\beta}_{1a}$ | $\hat{\beta}_{2a}$ | $\hat{\beta}_{2b}$ | $\Delta \hat{\beta}_{2a} - \Delta \hat{\beta}_{1a}$ |                 |
| 1976–1987                      | 197  | 0.907<br>15.366    | 0.632<br>11.692    | –0.274<br>–4.627          | 0.340<br>5.884     | 0.602<br>10.626    | 0.262<br>5.733                                      | 0.537<br>5.377  |
| 1988–2000                      | 269  | 1.011<br>23.314    | 0.691<br>11.554    | –0.364<br>–5.877          | 0.281<br>10.557    | 0.667<br>18.177    | 0.386<br>8.789                                      | 0.750<br>7.373  |
| 2001–2012                      | 214  | 0.951<br>13.407    | 0.691<br>9.456     | –0.260<br>–4.323          | 0.127<br>2.229     | 0.473<br>9.323     | 0.347<br>6.379                                      | 0.607<br>5.414  |
| 1976–2012                      | 680  | 0.962<br>29.394    | 0.657<br>18.504    | –0.305<br>–8.708          | 0.249<br>9.143     | 0.587<br>21.347    | 0.338<br>12.249                                     | 0.643<br>10.675 |

**Table 2**  
Stock splits.  
We estimate the univariate and bivariate regressions  
 $y_t = \alpha + \beta_1 x_{1t} + \varepsilon_t$   
 $y_t = \alpha + \beta_2 x_{2t} + \varepsilon_t$   
 $y_t = \alpha + \beta_{12} x_{1t} + \beta_{22} x_{2t} + \varepsilon_t$   
for a sample of two-for-one stock splits from 1971 through 2012. Our sample includes all ordinary common stock two-for-one splits with a presplit price of \$10 or greater during our sample period.  $x_{1t}$  and  $x_{2t}$  are return to a portfolio of high-priced stocks whose price belongs to [3p/4, 5p/4] and low-priced stocks with prices within [1p/4, 3p/4] at time  $t$ , where  $p$  is the presplit price before effective date of split. The pre-event (postevent) window is defined as the one year ending (beginning) one month before (after) the split date. Panel A reports the univariate regression results, and Panel B reports the bivariate regression results. In each cell, the first number is the mean, and the second number is the corresponding  $t$ -statistic, where standard errors are clustered by month.

| Panel A: Univariate regressions |      |                 |                  |                        |                 |                 |                        |                                               |
|---------------------------------|------|-----------------|------------------|------------------------|-----------------|-----------------|------------------------|-----------------------------------------------|
| High-priced group               |      |                 | Low-priced group |                        |                 | Diff. of diff.  |                        |                                               |
| Sample period                   | nobs | $\hat{\beta}_1$ | $\hat{\beta}_1$  | $\Delta \hat{\beta}_1$ | $\hat{\beta}_2$ | $\hat{\beta}_2$ | $\Delta \hat{\beta}_2$ | $\Delta \hat{\beta}_2 - \Delta \hat{\beta}_1$ |
| 1971–1990                       | 2350 | 0.736<br>48.771 | 0.929<br>54.709  | 0.193<br>17.138        | 0.847<br>49.059 | 1.043<br>60.555 | 0.196<br>18.554        | 0.002<br>0.443                                |
| 1991–2004                       | 2478 | 0.798<br>45.714 | 1.014<br>43.778  | 0.216<br>11.102        | 0.937<br>43.783 | 1.186<br>39.479 | 0.248<br>12.020        | 0.033<br>3.448                                |
| 1976–1987                       | 1867 | 0.729<br>40.663 | 0.919<br>45.952  | 0.190<br>14.620        | 0.847<br>40.182 | 1.036<br>50.304 | 0.189<br>15.136        | –0.001<br>–0.189                              |
| 1988–2000                       | 2383 | 0.796<br>44.584 | 1.001<br>42.256  | 0.205<br>10.371        | 0.968<br>46.228 | 1.206<br>39.782 | 0.237<br>11.141        | 0.032<br>3.256                                |
| 2001–2012                       | 794  | 0.932<br>30.602 | 1.141<br>36.926  | 0.209<br>8.553         | 0.887<br>28.309 | 1.084<br>36.203 | 0.197<br>8.518         | –0.012<br>–0.976                              |
| 1976–2012                       | 5044 | 0.792<br>63.604 | 0.993<br>66.954  | 0.200<br>17.930        | 0.910<br>64.844 | 1.124<br>63.436 | 0.213<br>18.142        | 0.013<br>2.281                                |

  

| Panel B: Bivariate regressions |      |                    |                    |                           |                    |                    |                                                     |                |
|--------------------------------|------|--------------------|--------------------|---------------------------|--------------------|--------------------|-----------------------------------------------------|----------------|
| High-priced group              |      |                    | Low-priced group   |                           |                    | Diff. of diff.     |                                                     |                |
| Sample period                  | nobs | $\hat{\beta}_{1a}$ | $\hat{\beta}_{1b}$ | $\Delta \hat{\beta}_{1a}$ | $\hat{\beta}_{2a}$ | $\hat{\beta}_{2b}$ | $\Delta \hat{\beta}_{2a} - \Delta \hat{\beta}_{1a}$ |                |
| 1971–1990                      | 2350 | –0.013<br>–0.724   | –0.043<br>–1.721   | –0.030<br>–1.011          | 0.865<br>38.366    | 1.085<br>41.082    | 0.220<br>7.235                                      | 0.250<br>4.231 |
| 1991–2004                      | 2478 | 0.041<br>1.379     | 0.003<br>0.092     | –0.038<br>–1.271          | 0.883<br>28.630    | 1.171<br>34.457    | 0.289<br>6.884                                      | 0.326<br>4.729 |
| 1976–1987                      | 1867 | –0.016<br>–0.938   | –0.068<br>–2.336   | –0.052<br>–1.539          | 0.863<br>34.684    | 1.101<br>35.398    | 0.238<br>6.884                                      | 0.290<br>4.311 |
| 1988–2000                      | 2383 | 0.001<br>0.027     | –0.035<br>–1.096   | –0.036<br>–1.160          | 0.951<br>29.898    | 1.224<br>37.527    | 0.273<br>6.124                                      | 0.309<br>4.229 |
| 2001–2012                      | 794  | 0.356<br>6.385     | 0.420<br>8.493     | 0.064<br>1.594            | 0.568<br>14.599    | 0.707<br>17.584    | 0.140<br>3.841                                      | 0.075<br>1.036 |
| 1976–2012                      | 5044 | 0.050<br>2.598     | 0.024<br>1.085     | –0.026<br>–1.271          | 0.858<br>43.961    | 1.097<br>48.520    | 0.239<br>9.363                                      | 0.265<br>5.956 |

### 5.3 Table 3: Numerical examples

Read:

- The base case has no coefficient changes.
- A change in the stock's fundamental beta creates a 0.192 increase in both univariate coefficients.
- A change in group idiosyncratic risk changes bivariate coefficients even without true excess comovement.
- A change in group beta can also generate bivariate coefficient changes.

**Economic message:** Table 3 shows why bivariate coefficients are fragile. The same apparent comovement patterns can arise from parameter instability unrelated to asset-class demand.

Table 3

Numerical examples.

We calculate the univariate and bivariate regression coefficients implied by the model in Section 2 for four numerical examples. In all cases, we assume no excess comovement, no correlation between idiosyncratic shocks at the group level, the same amount of idiosyncratic risk at the stock level, and the same amount of fundamental risk:

$$\sigma_{\epsilon_1} = \sigma_{\epsilon_2} = \sigma_{\epsilon_3} = \sigma_{\epsilon_4} = 0, \text{cov}(\epsilon_1, \epsilon_2) = 0, \sigma_{\eta_1} = \sigma_{\eta_2} = 1.73\%, \sigma_{\epsilon_1} = \sigma_{\epsilon_2} = 1\%.$$

The base case (top row) assumes perfect symmetry. The subsequent examples allow for parameter instability across the event, specifically (1) a change in the stock beta, (2) a change in idiosyncratic risk at the group level, and (3) a change in group beta. In each case, the deviations from the base case for both the input parameters and the regression coefficients are highlighted in bold.

Panel A: Inputs

| Case types                             | Fundamental loadings |       |       |            |       |            | Group idiosyncratic volatility |                       |                       |                       |                       |
|----------------------------------------|----------------------|-------|-------|------------|-------|------------|--------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                        | $b_1$                | $b_2$ | $b_3$ | $b_4$      | $b_5$ | $b_6$      | $\sigma_{\epsilon_1}$          | $\sigma_{\epsilon_2}$ | $\sigma_{\epsilon_3}$ | $\sigma_{\epsilon_4}$ | $\sigma_{\epsilon_5}$ |
| Base case                              | 1.0                  | 1.0   | 1.0   | 1.0        | 1.0   | 1.0        | 0.20%                          | 0.20%                 | 0.20%                 | 0.20%                 | 0.20%                 |
| (1) Change in stock beta               | 1.0                  | 1.0   | 1.0   | 1.0        | 1.0   | <b>1.2</b> | 0.20%                          | 0.20%                 | 0.20%                 | 0.20%                 | 0.20%                 |
| (2) Change in group idiosyncratic risk | 1.0                  | 1.0   | 1.0   | 1.0        | 1.0   | 1.0        | 0.20%                          | <b>0.24%</b>          | 0.20%                 | 0.20%                 | 0.20%                 |
| (3) Change in group beta               | 1.0                  | 1.0   | 1.0   | <b>0.8</b> | 1.0   | 1.0        | 0.20%                          | 0.20%                 | 0.20%                 | 0.20%                 | 0.20%                 |

Panel B: Regression coefficients

| Case types                             | Univariate  |              |           |                     |                     |                     |                     | Bivariate    |              |              |                           |                           |                           |                           |
|----------------------------------------|-------------|--------------|-----------|---------------------|---------------------|---------------------|---------------------|--------------|--------------|--------------|---------------------------|---------------------------|---------------------------|---------------------------|
|                                        | Coefficient |              |           | Change              |                     |                     |                     | Coefficient  |              |              | Change                    |                           |                           |                           |
|                                        | $\beta_1$   | $\beta_2$    | $\beta_3$ | $\beta_1 - \beta_2$ | $\beta_2 - \beta_3$ | $\beta_3 - \beta_4$ | $\beta_4 - \beta_5$ | $\beta_{12}$ | $\beta_{23}$ | $\beta_{34}$ | $\beta_{12} - \beta_{23}$ | $\beta_{23} - \beta_{34}$ | $\beta_{34} - \beta_{45}$ | $\beta_{45} - \beta_{56}$ |
| Base case                              | 0.962       | 0.962        | 0.962     | 0.000               | 0.000               | 0.000               | 0.000               | 0.490        | 0.490        | 0.490        | 0.000                     | 0.000                     | 0.000                     | 0.000                     |
| (1) Change in stock beta               | 0.962       | <b>1.154</b> | 0.962     | <b>1.154</b>        | <b>0.192</b>        | <b>0.192</b>        | 0.490               | <b>0.588</b> | 0.490        | <b>0.588</b> | <b>0.098</b>              | <b>0.098</b>              | <b>0.098</b>              | <b>0.098</b>              |
| (2) Change in group idiosyncratic risk | 0.962       | <b>0.946</b> | 0.962     | <b>-0.016</b>       | 0.000               | 0.000               | 0.490               | <b>0.490</b> | 0.490        | <b>0.577</b> | <b>-0.090</b>             | <b>0.086</b>              | <b>0.086</b>              | <b>0.086</b>              |
| (3) Change in group beta               | 0.962       | 0.962        | 0.962     | <b>1.176</b>        | 0.000               | <b>0.215</b>        | 0.490               | <b>0.595</b> | 0.490        | <b>0.476</b> | <b>0.105</b>              | <b>-0.014</b>             | <b>-0.014</b>             | <b>-0.014</b>             |

#### Table 4

Stock return performance of sample stocks.

For each stock in the sample (see Tables 1 and 2 for the sample description), we record the momentum portfolio in which it would fall based on its returns over the prior 12 months. The table reports the percentage of stocks in the S&P500 additions and stock splits samples that fall in each decile and the mean and median return on these stocks over the prior year.

| Decile        | Frequency (%)    |              |
|---------------|------------------|--------------|
|               | S&P500 additions | Stock splits |
| Losers        | 2.94             | 0.28         |
| 2             | 3.68             | 0.89         |
| 3             | 5.00             | 1.61         |
| 4             | 4.12             | 2.92         |
| 5             | 12.06            | 4.78         |
| 6             | 9.12             | 7.66         |
| 7             | 11.91            | 10.09        |
| 8             | 14.12            | 14.75        |
| 9             | 16.62            | 21.36        |
| Winners       | 20.44            | 35.66        |
| Mean return   | 41.6%            | 109.1%       |
| Median return | 25.0%            | 63.9%        |

If returns were independent of past returns, we would expect:

### 5.4 Table 4 and Table 5/Figure 1: Momentum and beta changes

Read:

- Table 4 shows that both S&P additions and stock splits are concentrated among past winners.
- For S&P additions, 20.44% are in the winner decile. For stock splits, 35.66% are in the winner decile.
- Table 5 shows that winner stocks experience rising betas around the momentum formation period.
- Figure 1 visualizes this pattern: winner betas rise sharply into and after formation, while loser betas drift down.

**Economic message:** The event samples are not neutral samples. They contain winner stocks, and winner stocks naturally have increasing betas. This explains much of the original comovement evidence.



Table 5

Beta changes and momentum.

At the end of each June from 1976 through 2011, stocks with a price of at least \$10 that do not fall into the bottom size decile of NYSE stocks are assigned into ten momentum deciles based on their cumulative returns over the preceding 252 days. We estimate betas for each stock based on a rolling window of 252 days from two years before formation of momentum portfolios through two years after formation, and compare beta changes for both the top and bottom two momentum portfolios. Thus, betas for years -2 and -1 are estimated over rolling windows ending 504 and 252 trading days before portfolio formation, respectively. Postmomentum portfolio formation years allow for a 21-trading day skip, and are estimated over 252 days ending 273 and 525 trading days after portfolio formation. In each cell, the first number is the time series average of the mean, and the second number is the corresponding *t*-statistic.

| Momentum decile  | Year -2         | Year -1         | Year 0          | Year 1          | Year 2          | Year 0 - Year -2 | Year 0 - Year -1 | Year 1 - Year 0  | Year 2 - Year 0  |
|------------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|------------------|
| 10 (Winners)     | 0.964<br>21.865 | 0.976<br>20.588 | 1.143<br>21.313 | 1.271<br>21.163 | 1.166<br>25.997 | 0.179<br>4.139   | 0.167<br>4.891   | 0.128<br>3.217   | 0.023<br>0.553   |
| 9                | 0.918<br>21.954 | 0.916<br>21.109 | 0.981<br>21.917 | 1.038<br>23.529 | 0.994<br>27.510 | 0.063<br>2.161   | 0.065<br>2.695   | 0.057<br>2.194   | 0.013<br>0.448   |
| Middle 6 deciles | 0.818<br>25.686 | 0.829<br>28.510 | 0.832<br>27.432 | 0.834<br>26.003 | 0.841<br>25.132 | 0.014<br>0.642   | 0.014<br>0.221   | 0.002<br>0.150   | 0.009<br>0.438   |
| 2                | 0.903<br>28.188 | 0.925<br>28.429 | 0.917<br>23.377 | 0.876<br>20.314 | 0.888<br>20.436 | 0.014<br>0.579   | -0.007<br>-0.373 | -0.042<br>-2.546 | -0.029<br>-1.251 |
| 1 (Losers)       | 1.047<br>31.351 | 1.094<br>32.901 | 1.092<br>21.759 | 1.031<br>20.707 | 1.015<br>21.423 | 0.045<br>1.229   | -0.003<br>-0.091 | -0.061<br>-1.977 | -0.077<br>-2.219 |



## 5.5 Table 6: S&P 500 additions with matched samples and Dimson adjustments

Read:

- The table compares the original additions sample, matched stocks, and sample-minus-match beta changes.
- Matched stocks display similar beta changes to addition stocks.
- After subtracting matched-stock changes, most apparent excess comovement is small or statistically insignificant.
- Dimson adjustment further weakens the evidence by addressing nonsynchronous trading effects.

Table 6

S&amp;P additions with matched-sample and Dimson adjustments.

We estimate the univariate and bivariate Dimson (1979) regressions for a sample of stocks that are added to the S&P500 index from 1976 through 2012 and for a portfolio of matched firms. The pre-event estimation period covers a one-year window ending at the end of the month preceding announcement, while the postevent period covers the one-year window starting the month after the effective date of index change.  $x_{it}$  is return to non-S&P500 index at time  $t$ , while  $x_{it}$  is return to the S&P500 index at time  $t$ . The match firm for each addition is identified as the one with closest momentum from the same size decile as the addition firms. The Dimson beta is defined as a simple sum of the lag, concurrent, and lead coefficients from the following regressions with two leads and lags. In each cell, the first number is the mean, and the second number is the corresponding *t*-statistic, where standard errors are clustered by month.

$$y_t = \alpha + \sum_{s=-2}^2 \beta_1^s x_{t+s} + \varepsilon_t$$

$$y_t = \alpha + \sum_{s=-2}^2 \beta_2^s x_{t+s} + \varepsilon_t$$

$$y_t = \alpha + \sum_{s=-2}^2 \beta_1^s x_{t+s} + \sum_{s=-2}^2 \beta_2^s x_{t+s} + \varepsilon_t$$

| Panel A: Univariate regressions |               |      |                    |                    |                          |                    |                    |                          |                                                   |
|---------------------------------|---------------|------|--------------------|--------------------|--------------------------|--------------------|--------------------|--------------------------|---------------------------------------------------|
|                                 | Sample period | nobs | Non-S&P500 group   |                    |                          | S&P500             |                    |                          | Diff. of diff.                                    |
|                                 |               |      | $\hat{\beta}_1$    | $\hat{\beta}_1$    | $\Delta\hat{\beta}_1$    | $\hat{\beta}_2$    | $\hat{\beta}_2$    | $\Delta\hat{\beta}_2$    | $\Delta\hat{\beta}_2 - \Delta\hat{\beta}_1$       |
| Sample                          | 1976-1987     | 187  | 1.190<br>28.734    | 1.272<br>28.294    | 0.081<br>2.017           | 1.156<br>26.699    | 1.190<br>29.071    | 0.033<br>0.820           | -0.048<br>-2.096                                  |
|                                 | 1988-2000     | 245  | 1.193<br>27.716    | 1.142<br>23.419    | -0.051<br>-0.919         | 1.161<br>29.551    | 1.239<br>25.360    | 0.078<br>1.527           | 0.129<br>2.549                                    |
|                                 | 2001-2012     | 203  | 1.007<br>26.383    | 1.020<br>25.779    | 0.013<br>0.348           | 1.187<br>29.117    | 1.168<br>27.880    | -0.020<br>-0.483         | -0.033<br>-1.187                                  |
|                                 | 1976-2012     | 635  | 1.133<br>46.873    | 1.141<br>42.120    | 0.008<br>0.312           | 1.168<br>49.382    | 1.202<br>45.861    | 0.034<br>1.260           | 0.025<br>1.093                                    |
| Match                           | 1976-1987     | 187  | 1.060<br>27.537    | 1.066<br>26.633    | 0.006<br>0.143           | 0.985<br>27.133    | 0.989<br>23.089    | 0.004<br>0.103           | -0.002<br>-0.085                                  |
|                                 | 1988-2000     | 245  | 1.103<br>23.047    | 1.071<br>19.611    | -0.031<br>-0.493         | 1.087<br>25.860    | 1.167<br>17.404    | 0.080<br>1.156           | 0.111<br>2.232                                    |
|                                 | 2001-2012     | 203  | 0.976<br>21.149    | 0.994<br>24.104    | 0.017<br>0.437           | 1.138<br>22.733    | 1.132<br>23.749    | -0.006<br>-0.126         | -0.023<br>-0.877                                  |
|                                 | 1976-2012     | 635  | 1.050<br>40.310    | 1.045<br>38.185    | -0.005<br>-0.160         | 1.073<br>42.135    | 1.103<br>33.184    | 0.030<br>0.916           | 0.035<br>1.552                                    |
| Sample minus match              | 1976-1987     | 187  | 0.130<br>2.721     | 0.206<br>3.783     | 0.076<br>1.321           | 0.171<br>3.965     | 0.201<br>3.752     | 0.030<br>0.550           | -0.046<br>-1.998                                  |
|                                 | 1988-2000     | 245  | 0.090<br>1.821     | 0.070<br>1.363     | -0.020<br>-0.392         | 0.074<br>1.481     | 0.072<br>1.445     | -0.002<br>-0.029         | 0.018<br>0.601                                    |
|                                 | 2001-2012     | 203  | 0.031<br>0.537     | 0.026<br>0.506     | -0.004<br>-0.094         | 0.050<br>0.806     | 0.036<br>0.660     | -0.014<br>-0.247         | -0.009<br>-0.458                                  |
|                                 | 1976-2012     | 635  | 0.083<br>2.764     | 0.096<br>3.097     | 0.013<br>0.447           | 0.095<br>3.104     | 0.099<br>3.179     | 0.004<br>0.121           | -0.009<br>-0.631                                  |
| Panel B: Bivariate regressions  |               |      |                    |                    |                          |                    |                    |                          |                                                   |
|                                 | Sample period | nobs | Non-S&P500 group   |                    |                          | S&P500             |                    |                          | Diff. of diff.                                    |
|                                 |               |      | $\hat{\beta}_{1b}$ | $\hat{\beta}_{1b}$ | $\Delta\hat{\beta}_{1b}$ | $\hat{\beta}_{2b}$ | $\hat{\beta}_{2b}$ | $\Delta\hat{\beta}_{2b}$ | $\Delta\hat{\beta}_{2b} - \Delta\hat{\beta}_{1b}$ |
| Sample                          | 1976-1987     | 187  | 0.756<br>10.061    | 0.809<br>10.040    | 0.053<br>0.489           | 0.462<br>7.552     | 0.452<br>5.819     | -0.010<br>-0.100         | -0.063<br>-0.307                                  |
|                                 | 1988-2000     | 245  | 0.803<br>14.017    | 0.693<br>9.300     | -0.110<br>-1.271         | 0.438<br>8.225     | 0.562<br>8.152     | 0.123<br>1.579           | 0.233<br>1.473                                    |
|                                 | 2001-2012     | 203  | 0.901<br>10.262    | 0.791<br>7.895     | -0.109<br>-1.220         | 0.148<br>1.765     | 0.270<br>2.817     | 0.123<br>1.219           | 0.232<br>1.241                                    |
|                                 | 1976-2012     | 635  | 0.820<br>19.467    | 0.759<br>15.474    | -0.062<br>-1.145         | 0.352<br>8.889     | 0.436<br>9.269     | 0.084<br>1.583           | 0.146<br>1.399                                    |
| Match                           | 1976-1987     | 187  | 0.844<br>12.157    | 0.803<br>12.962    | -0.041<br>-0.491         | 0.221<br>3.462     | 0.265<br>4.522     | 0.044<br>0.558           | 0.085<br>0.542                                    |
|                                 | 1988-2000     | 245  | 0.800<br>12.115    | 0.785<br>10.199    | -0.015<br>-0.177         | 0.368<br>6.708     | 0.406<br>5.661     | 0.038<br>0.452           | 0.053<br>0.332                                    |

(continued on next page)

Table 6 (continued)

| Panel B: Bivariate regressions |               |      |                    |                    |                          |                    |                    |                          |                                                   |
|--------------------------------|---------------|------|--------------------|--------------------|--------------------------|--------------------|--------------------|--------------------------|---------------------------------------------------|
|                                | Sample period | nobs | Non-S&P500 group   |                    |                          | S&P500             |                    |                          | Diff. of diff.                                    |
|                                |               |      | $\hat{\beta}_{1b}$ | $\hat{\beta}_{1b}$ | $\Delta\hat{\beta}_{1b}$ | $\hat{\beta}_{2b}$ | $\hat{\beta}_{2b}$ | $\Delta\hat{\beta}_{2b}$ | $\Delta\hat{\beta}_{2b} - \Delta\hat{\beta}_{1b}$ |
| Sample minus match             | 2001-2012     | 203  | 0.893<br>8.893     | 0.808<br>7.155     | -0.086<br>-0.819         | 0.107<br>1.087     | 0.209<br>1.918     | 0.102<br>0.905           | 0.188<br>0.881                                    |
|                                | 1976-2012     | 635  | 0.843<br>18.253    | 0.798<br>15.937    | -0.045<br>-0.856         | 0.241<br>2.467     | 0.301<br>6.242     | 0.060<br>1.122           | 0.105<br>1.027                                    |
|                                | 1976-1987     | 187  | -0.087<br>-0.816   | 0.006<br>0.059     | 0.006<br>0.648           | 0.093<br>2.806     | 0.241<br>1.886     | 0.187<br>-0.415          | -0.054<br>-0.550                                  |
|                                | 1988-2000     | 245  | 0.003<br>0.040     | -0.092<br>-1.035   | -0.095<br>-1.019         | 0.070<br>0.942     | 0.156<br>1.955     | 0.086<br>0.930           | 0.181<br>1.014                                    |
| Sample minus match             | 2001-2012     | 203  | 0.008<br>0.067     | -0.016<br>-0.111   | -0.024<br>-0.188         | 0.041<br>0.392     | 0.061<br>0.432     | 0.020<br>0.147           | 0.044<br>0.170                                    |
|                                | 1976-2012     | 635  | 0.003<br>0.003     | 0.003<br>0.003     | 0.003<br>0.003           | 0.003<br>0.003     | 0.003<br>0.003     | 0.003<br>0.003           | 0.003<br>0.003                                    |

**Economic message:** Table 6 is the main rebuttal for S&P additions. Once size/momentum matching and Dimson adjustment are used, excess comovement largely disappears.

## 5.6 Table 7: Stock splits with matched samples and Dimson adjustments

Read:

- The table compares split stocks with matched stocks selected on size and momentum.
- The sample and matched stocks exhibit similar coefficient changes.
- The sample-minus-match differences are generally small.
- There is little evidence that stock splits create economically meaningful excess comovement with low-priced stocks.

**Economic message:** Table 7 shows that the stock-split evidence is explained mainly by the momentum and beta characteristics of splitting firms, not by price-category asset-class demand.

Table 7

Stock splits with matched-sample and Dimson adjustments.

We estimate the univariate and bivariate Dimson (1979) regressions for a sample of two-for-one stock splits from 1976 through 2012. Our sample includes all ordinary common stock two-for-one splits with a presplit price of \$10 or greater during our sample period.  $x_{it}$  and  $x_{jt}$  are return to a portfolio of high-priced stocks whose price belongs to [3p/4, 5p/4] and low-priced stocks with prices within [1p/4, 3p/4] at time  $t$ , where  $p$  is the presplit price before effective date of split. The pre-event (postevent) window is defined as the one year ending (beginning) one month before (after) the split date. The Dimson beta is defined as a simple sum of the lag, concurrent, and lead coefficients from the following regressions with two leads and lags. In each cell, the first number is the mean, and the second number is the corresponding  $t$ -statistic, where standard errors are clustered by month.

$$y_t = \alpha + \sum_{k=-2}^2 \beta_1^k x_{1,t+k} + \varepsilon_t$$

$$y_t = \alpha + \sum_{k=-2}^2 \beta_2^k x_{2,t+k} + \varepsilon_t$$

$$y_t = \alpha + \sum_{k=-2}^2 \beta_1^k x_{1,t+k} + \sum_{k=-2}^2 \beta_2^k x_{2,t+k} + \varepsilon_t$$

| Panel A: Univariate regressions |               |      |                   |                 |                 |                  |                 |                 |                  |
|---------------------------------|---------------|------|-------------------|-----------------|-----------------|------------------|-----------------|-----------------|------------------|
|                                 | Sample period | nobs | High-priced group |                 |                 | Low-priced group |                 |                 | Diff. of diff.   |
|                                 |               |      | $\beta_1$         | $\hat{\beta}_1$ | $\Delta\beta_1$ | $\beta_2$        | $\hat{\beta}_2$ | $\Delta\beta_2$ |                  |
| Sample                          | 1976-1987     | 1606 | 0.924<br>55.306   | 1.120<br>42.904 | 0.197<br>8.486  | 0.981<br>55.824  | 1.154<br>51.170 | 0.172<br>8.603  | -0.024<br>-2.271 |
|                                 | 1988-2000     | 2097 | 0.963<br>42.614   | 1.132<br>41.875 | 0.168<br>41.875 | 1.056<br>45.824  | 1.267<br>40.457 | 0.211<br>7.029  | 0.043<br>2.834   |
|                                 | 2001-2012     | 727  | 1.042<br>29.491   | 1.222<br>37.391 | 0.180<br>5.181  | 0.961<br>26.458  | 1.135<br>34.593 | 0.174<br>5.621  | -0.006<br>-0.411 |
|                                 | 1976-2012     | 4430 | 0.962<br>69.502   | 1.142<br>67.815 | 0.181<br>10.295 | 1.014<br>72.336  | 1.204<br>66.694 | 0.191<br>11.367 | 0.010<br>1.172   |
|                                 | 1976-1987     | 1606 | 0.858<br>48.739   | 0.957<br>42.268 | 0.099<br>5.078  | 0.901<br>50.803  | 0.981<br>48.675 | 0.080<br>4.855  | -0.019<br>-2.349 |
| Match                           | 1988-2000     | 2097 | 0.847<br>46.451   | 0.943<br>40.139 | 0.095<br>3.492  | 0.926<br>49.136  | 1.046<br>33.341 | 0.120<br>4.236  | 0.025<br>2.112   |
|                                 | 2001-2012     | 727  | 0.991<br>30.774   | 1.116<br>42.113 | 0.125<br>4.538  | 0.917<br>29.015  | 1.029<br>37.387 | 0.112<br>4.852  | -0.013<br>-1.031 |
|                                 | 1976-2012     | 4430 | 0.875<br>70.200   | 0.976<br>66.242 | 0.102<br>6.620  | 0.915<br>75.350  | 1.020<br>59.093 | 0.104<br>6.856  | 0.003<br>0.375   |
|                                 | 1976-1987     | 1606 | 0.066<br>4.459    | 0.163<br>8.755  | 0.098<br>4.774  | 0.080<br>5.694   | 0.173<br>7.490  | 0.093<br>4.620  | -0.005<br>-0.688 |
|                                 | 1988-2000     | 2097 | 0.116<br>6.757    | 0.189<br>9.134  | 0.073<br>3.493  | 0.131<br>7.490   | 0.222<br>10.703 | 0.091<br>4.638  | 0.018<br>1.619   |
| Sample minus match              | 2001-2012     | 727  | 0.050<br>1.823    | 0.105<br>3.372  | 0.055<br>1.790  | 0.045<br>1.727   | 0.106<br>3.437  | 0.062<br>2.099  | 0.007<br>0.582   |
|                                 | 1976-2012     | 4430 | 0.087<br>8.081    | 0.166<br>12.819 | 0.079<br>5.911  | 0.098<br>9.204   | 0.185<br>14.350 | 0.087<br>6.826  | 0.008<br>1.249   |

(continued on next page)

Table 7 (continued)

| Panel B: Bivariate regressions |               |      |                   |                    |                    |                  |                    |                    |                |
|--------------------------------|---------------|------|-------------------|--------------------|--------------------|------------------|--------------------|--------------------|----------------|
|                                | Sample period | nobs | High-priced group |                    |                    | Low-priced group |                    |                    | Diff. of diff. |
|                                |               |      | $\beta_{1b}$      | $\hat{\beta}_{1b}$ | $\Delta\beta_{1b}$ | $\beta_{2b}$     | $\hat{\beta}_{2b}$ | $\Delta\beta_{2b}$ |                |
| Sample                         | 1976-1987     | 1606 | 0.262<br>7.176    | 0.152<br>3.754     | -0.109<br>-1.915   | 0.722<br>20.178  | 0.993<br>25.398    | 0.272<br>4.645     | 0.381<br>3.349 |
|                                | 1988-2000     | 2097 | 0.177<br>4.073    | 0.008<br>0.161     | -0.170<br>-3.172   | 0.866<br>18.434  | 1.238<br>27.571    | 0.372<br>5.726     | 0.542<br>4.741 |
|                                | 2001-2012     | 727  | 0.471<br>7.325    | 0.464<br>8.411     | -0.007<br>-0.101   | 0.554<br>11.453  | 0.731<br>13.473    | 0.176<br>3.055     | 0.183<br>1.513 |
|                                | 1976-2012     | 4430 | 0.256<br>9.363    | 0.135<br>4.514     | -0.121<br>-3.492   | 0.762<br>27.557  | 1.066<br>36.602    | 0.304<br>7.847     | 0.425<br>5.973 |
|                                | 1976-1987     | 1606 | 0.317<br>9.210    | 0.172<br>4.637     | -0.144<br>-2.832   | 0.588<br>17.657  | 0.810<br>20.767    | 0.221<br>4.132     | 0.366<br>3.549 |
| Match                          | 1988-2000     | 2097 | 0.226<br>6.847    | 0.109<br>3.114     | -0.117<br>-2.725   | 0.698<br>21.726  | 0.941<br>24.571    | 0.243<br>4.906     | 0.360<br>4.129 |
|                                | 2001-2012     | 727  | 0.455<br>8.797    | 0.482<br>9.691     | 0.027<br>0.432     | 0.518<br>12.138  | 0.616<br>14.502    | 0.098<br>1.893     | 0.072<br>0.651 |
|                                | 1976-2012     | 4430 | 0.296<br>13.273   | 0.193<br>7.960     | -0.103<br>-3.482   | 0.628<br>29.628  | 0.840<br>33.409    | 0.211<br>6.631     | 0.315<br>5.314 |
|                                | 1976-1987     | 1606 | -0.055<br>-1.279  | -0.020<br>-0.508   | 0.035<br>0.603     | 0.133<br>3.086   | 0.184<br>4.745     | 0.050<br>0.841     | 0.015<br>0.134 |
|                                | 1988-2000     | 2097 | -0.049<br>-1.240  | -0.101<br>-2.548   | -0.053<br>-0.944   | 0.168<br>4.067   | 0.298<br>7.071     | 0.130<br>2.347     | 0.182<br>1.672 |
| Sample minus match             | 2001-2012     | 727  | 0.015<br>0.231    | -0.018<br>-0.268   | -0.033<br>-0.470   | 0.037<br>0.604   | 0.115<br>1.745     | 0.078<br>1.116     | 0.112<br>0.813 |
|                                | 1976-2012     | 4430 | -0.040<br>-1.517  | -0.058<br>-2.228   | -0.018<br>-0.496   | 0.134<br>4.955   | 0.226<br>8.443     | 0.092<br>2.575     | 0.110<br>1.568 |

## 6 Short summary

- The paper revisits excess comovement evidence from S&P 500 additions and stock splits.
- The prior literature relies heavily on bivariate regressions.
- The authors show that bivariate coefficients are sensitive to fundamental beta changes, group risk changes, and other parameter instability.
- Univariate regressions on the old group are important and often do not support the excess-comovement story.
- S&P additions and stock split samples are heavily populated by past winner stocks.
- Winner stocks have rising betas around formation/event dates.
- Matched stocks with similar size and momentum exhibit similar beta changes even though they do not change groups.
- Dimson adjustment is important for S&P additions because of nonsynchronous trading and microstructure effects.
- After matched-sample and Dimson adjustments, almost all evidence of excess comovement disappears.
- The paper's central lesson is that apparent nonfundamental comovement can be an artifact of time-varying betas and sample selection.

## **7 Conclusion**

### **7.1 Core conclusion**

Chen, Singal, and Whitelaw argue that the empirical evidence for excess comovement around S&P 500 additions and stock splits is much weaker than previously believed. The bivariate regression evidence is fragile, and matched-sample tests show that similar beta changes occur for nonevent stocks with similar size and momentum.

### **7.2 Economic intuition**

Stocks that join the S&P 500 or split are often past winners. Winner stocks experience rising betas, so their returns begin to load more on broad market or group portfolios even without any nonfundamental asset-class demand. If this beta change is not controlled for, it can look like excess comovement.

### **7.3 Takeaway**

The paper is a methodological warning: regression coefficient changes are not automatically evidence of nonfundamental comovement. To measure economically meaningful asset-class demand effects, one must separate true group-specific shocks from changes in fundamental betas and stock selection into high-momentum samples.